DATA ANALYST 3: ADVANCED SQL

Student Tuition & Demographic Analysis

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01 | Introduction

As a Data Analyst, I wanted to explore whether factors such as gender, ethnicity, education level, family income, and parental education were associated with differences in student tuition and wages.

The analysis used four datasets:

- country_info
- student_academic_info
- student_family_details
- student_personal_details

I used INNER JOINs to combine the four datasets through their common ID fields, creating a consolidated analytical table that allowed student academic, personal, family, and geographic information to be examined together.

02 | Analysis Questions

The analysis was designed to explore potential relationships between demographic, socioeconomic, geographic, and family-education factors within the student dataset.

- How does tuition vary by gender and ethnicity?
- Does parental college attendance vary across demographic groups?
- How does family income vary by gender and ethnicity?
- Does geographic region relate to average wages?
- What additional information would be needed to better understand differences in student tuition?

03 | SQL & Data Integration

The four datasets were joined using a common ID to create the consolidated table **reports_student_colleges**. The SQL below reflects the original analysis while correcting the grouping logic used for parental college-attendance queries.

Create Consolidated Student Table

```
CREATE TABLE reports_student_colleges AS
SELECT *
FROM country_info
INNER JOIN student_academic_info
ON country_info.id = student_academic_info.id
INNER JOIN student_family_details
ON country_info.id = student_family_details.id
INNER JOIN student_personal_details
ON country_info.id = student_personal_details.id;
```

Mother's College Attendance - Female Students

```
SELECT ethnicity, mcollege, COUNT(*) AS student_count
FROM reports_student_colleges
WHERE gender = 'female'
GROUP BY ethnicity, mcollege
ORDER BY ethnicity, mcollege;
```

Mother's College Attendance - Male Students

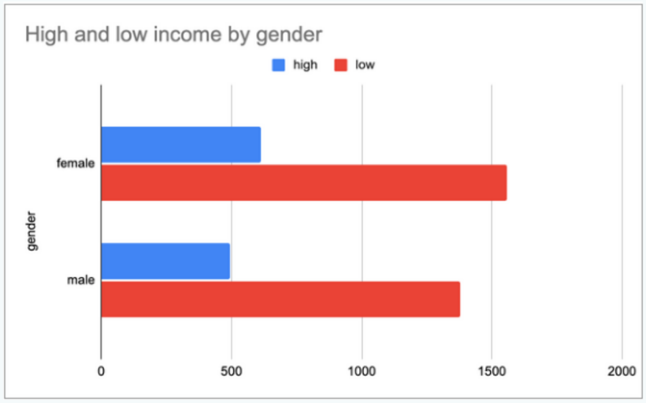
```
SELECT ethnicity, mcollege, COUNT(*) AS student_count
FROM reports_student_colleges
WHERE gender = 'male'
GROUP BY ethnicity, mcollege
ORDER BY ethnicity, mcollege;
```

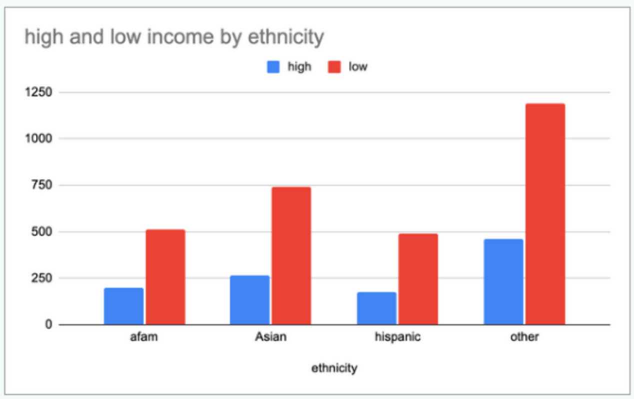
Additional portfolio-ready queries supporting the tuition, parental education, income, and regional wage analyses are included in the accompanying GitHub SQL file.

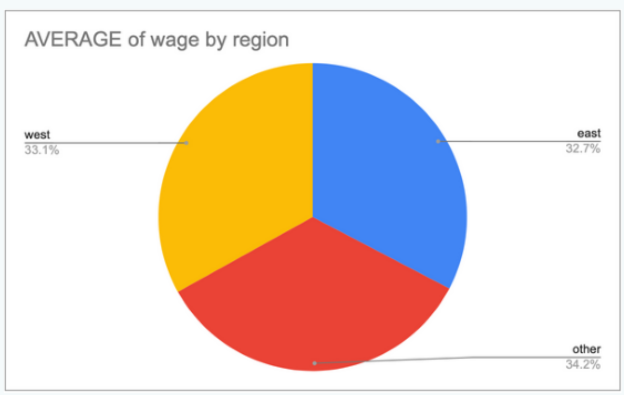
04 | Visualizations

The original project visualizations are preserved below. They summarize average tuition, parental college attendance, income patterns, and average wages by region.

Visualizations cont'd







06

05 | Findings

The analysis found that Asian female students had the highest average tuition within the dataset, while Hispanic male students also had comparatively high average tuition. African American male and female students had lower average tuition among the groups analyzed.

These differences may be associated with factors not fully represented in the available data, such as degree program, financial aid, or other socioeconomic characteristics. Additional data would be needed to determine what factors contribute to the observed tuition differences.

Parental college attendance also varied across demographic groups. However, interpretation of the "Other" ethnicity category is limited because it may combine multiple demographic groups.

Average wages were relatively similar across the East, West, and "Other" regional categories.

Recommended next steps

- Examine students' degree programs to determine whether program selection may help explain tuition differences.
- Investigate the relationship between family income, financial aid, and tuition.
- Break down the broad "Other" ethnicity category into more specific demographic groups.
- Provide greater detail about the geographic areas included in the "Other" region category.

06 | Conclusion

The analysis identified differences in tuition, parental education, income, and wages across several demographic and geographic groups. Asian female students had the highest average tuition within the dataset, while parental college attendance also varied across student groups.

However, the available data is not sufficient to determine why these differences exist. Additional information about degree programs, financial aid, socioeconomic factors, and the demographic groups represented within the "Other" categories would support a more comprehensive analysis.

Portfolio note: This updated edition preserves the original analytical scope and visualizations while refining grammar, terminology, SQL grouping logic, and the distinction between observed associations and causal conclusions.